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Vegetative and reproductive stage lighting interactions on flower yield, water use efficiency, terpenes, and cannabinoids of *Cannabis sativa*

C. E. Collado✉ & R. Hernández✉

Cannabis is among the most water-demanding crops, facing ongoing expansion and water use regulations. This study evaluated the effects of greenhouse sunlight plus increased LED supplemental lighting on flower yield, water use (WU), and efficiency (WUE), as well as flower partitioning, cannabinoids, terpenes, and leaf gas exchange in *Cannabis sativa* 'Suver Haze'. The supplemental lighting programs applied during the vegetative and flowering stages were: (1) Static LED levels (PPFD: 150, 300, 500, or 700 µmol m^{-2} s^{-1} for 72 days) and (2) Dynamic LED levels (PPFD: 150, 300, or 500 for 28 days, followed by 700 µmol m^{-2} s^{-1} for 44 days). Flower yield and crop WUE increased linearly with the increase of Dynamic and Static lighting. For instance, a 4.7-fold Static lighting increase caused a 4.1-fold increase in flower yield and reduced the evapotranspiration per gram of flower by 37%. Furthermore, plants in Dynamic lighting produced up to 10.4% more flowers and 24.8% higher WUE than plants in Static lighting at the same cumulative lighting. Higher leaf photosynthetic rate, WUE, and lower stomatal conductance due to higher light intensity supported the crop results. Cannabinoid and terpene changes were small and complex, with terpene concentration affected by the light program and light level. In conclusion, supplemental lighting substantially enhanced production and WUE, particularly when higher light was provided during flowering.

Keywords Hemp, Sustainable production, Indoor production, Water savings, Secondary metabolites, Supplemental lighting, Terpenoids, Water footprint

The development of cannabis production has experienced persistent expansion due to ongoing legalization across several states in the US[1]. However, recent data indicate that both field and greenhouse cannabis cultivations are among the most water-demanding crops[2]. California, the primary production state in the US, already allocates between 70 and 80% of surface freshwater and groundwater for agriculture[2]. Furthermore, the US applied 42% of its freshwater for irrigation in 2015[3]. With growing concerns about water infrastructure, demand, and shortages[4], it is critical to understand and reduce the water footprint of cannabis and other crops for feasibility, greenhouse water infrastructure planning, and sustainability. Nevertheless, data on this topic remain scarce and variable[5,6].

Cannabis is a high-light-demanding crop[7]. While greenhouse cultivation exploits solar light to reduce production costs[8], solar radiation in the greenhouse may be insufficient to achieve the desired flower quality and yield. Several studies have evaluated the impact of electrical lighting (HPS or LED as sole source lighting) during the flowering period only, providing insights into flower growth under those conditions[9–12]. Still, greenhouse lighting studies are limited[13] and have not been conducted during the vegetative or vegetative plus flowering periods for flower production. Moreover, during the vegetative stage, the cumulative photosynthetic photon flux density (ΣPPFD), solar plus LED lighting, has been positively correlated with plant size, branching, and the number of nodes[7]; and since cannabis inflorescences originate from nodes[14], greater node development could lead to increased flower production. Yet, maximum yield may still occur at smaller plant sizes or lower node counts due to resource constraints[15]. Ultimately, the influence of environmental conditions and plant traits during and at the end of the vegetative stage on final flower yield remains largely unknown.

Department of Horticultural Science, North Carolina State University, Raleigh, NC, USA. ✉email: cecollad@ncsu.edu; rhernan4@ncsu.edu

Increased lighting also raises the crop water demand. Nonetheless, we previously reported how increasing the daily light integral (DLI) from 18 to 52 mol m^{-2} d^{-1} increased water-use-efficiency in cannabis and saved as much as 35% of the water usage during the vegetative stage[7]. Likewise, lettuce (*Lactuca sativa* L.) water losses from transpiration were reduced by 46% with a DLI increase from 8 to 22 mol m^{-2} d^{-1} on the same biomass basis[16]. Collado et al.[7] reviewed the primary physiological mechanisms and their regulations associated with plant growth and water regulation (i.e., leaf photosynthesis, transpiration, and stomatal conductance), explaining water savings due to increased lighting. However, the literature lacks measured data on cannabis flower production, specifically regarding water usage from planting to harvest, as well as the impact of lighting and other factors on water use efficiency.

Cannabis is recognized primarily for its high concentration of cannabinoids in female flowers, which are often used for their effects on the human body. Terpenes, in addition to providing the cannabis aroma, have potential medical uses and interaction effects with cannabinoids. Terpenes show cannabinoid tetrad activities, influence cannabinoid receptors, and may enhance cannabinoid therapeutic effects[17–19]. Both metabolites are produced and stored in glandular trichomes; however, terpene and cannabinoid synthesis can vary independently[20].

Cannabinoid and terpene concentrations can decrease from the top to the lower flowers[11,21–23] and from older to younger flowers, while the terpene composition also changes with flower and trichome maturity[20,24,25]. Compared to the upper flowers on the plant's main stem, those on lower branches seem more likely to grow under limited photoassimilates and vary in trichome or flower maturity. Despite, or perhaps due to, the potential decline in secondary metabolites in lower flowers, the effects of light have predominantly been evaluated on apical or upper flowers[10,12,26,27], potentially overlooking metabolite concentration reduction or increments in other flowers. For example, compared to top flowers, metabolite concentrations in middle and low flowers dropped by ~ 5 and 11% in terpenes, ~ 23 and 82% in cannabinoids[23], and ~ 17 and 25% in total cannabidiol (CBD)[21] but not always[11]. Conversely, literature reports indicate that top flowers exhibit stable terpene and cannabinoid concentrations or increased terpene concentrations with additional lighting[10,12,26,27]. Therefore, to fully understand flower production and quality under varying production conditions, it is crucial to characterize the effects of lighting on flower biomass distribution along the plant and the production of cannabinoids and terpenes, particularly in less-studied flowers.

This research evaluated the impact of greenhouse solar plus LED light levels during the vegetative and flowering stages (Objective 1) and before the flower growth period (Objective 2) on crop flower yield, quality, and water use and efficiency. Plant parameters such as leaf photosynthesis, stomatal conductance, and transpiration, as well as cannabinoid and terpene concentrations and compositions in lower-node flowers, were also assessed. The minimum to maximum common greenhouse light levels were selected, spanning recommended values for vegetable, fruit, and high-light crops, such as cannabis and greenhouse corn production for speed breeding[7].

We hypothesized that increasing cumulative lighting (ΣPPFD) would improve leaf and crop growth and resource efficiency parameters, i.e., leaf photosynthesis (*A*), transpiration (*E*), WUE (*A*/*E*), flower mass yield, crop water use (WU) and efficiency (WUE: flowers per liter of water), and concentrations of cannabinoids and terpenes. However, we also hypothesized that reducing supplemental light levels (PPFD) before the flowering growth period would have a low impact on the final flower yield, increasing the light use efficiency (flowers per mole of light) and the crop WUE.

Results and discussion

Light effects on flower yield and crop water usage

Flower yield

Light is critical for generating new plant structures such as flowers. The light intensity (PPFD, or photosynthetic photons per square meter per second) and accumulated lighting (ΣPPFD, or accumulated photons per square meter over the production cycle) can affect crop responses[7]. Therefore, this greenhouse study quantified the impact of cumulative light, solar plus electrical (ΣPPFD: 1065 ± 4 to 2978 ± 85 mol m^{-2}), and two supplemental light programs (LP: Static or Dynamic PPFDs), spanning the vegetative and reproductive cycles (Table 1). Increased lighting improved flower growth linearly in both the Static and Dynamic LP at a rate of 0.215 g (95%

Lighting Program (LP)	LED PPFD [y] µmol m^{-2} s^{-1}	N° of plots or subplots	ΣPPFD [z] mol m^{-2}	Avg. DLI [z] mol m^{-2} d^{-1}	Light source [z]	
					Sun	LED
Static	150 / 150	2	1065 ± 4	14.6	50%	50%
Static	300 / 300	2	1598 ± 1	21.9	34%	66%
Dynamic	150 / 700	3	2109 ± 53	28.9	26%	74%
Static	500 / 500	2	2312 ± 8	31.7	24%	76%
Dynamic	300 / 700	3	2385 ± 47	32.7	24%	76%
Dynamic	500 / 700	3	2694 ± 40	36.9	21%	79%
Static	700 / 700	3	2978 ± 85	40.8	17%	83%

Table 1. Seven lighting treatments ranging from lower to higher cumulative photosynthetic photon flux densities (ΣPPFD). [y] LED supplemental lighting treatments from day 1 to 29 and 29 to 73 after planting. [z] the average total, daily, and proportion of LED and solar lighting from two and three replications (n° of plots or subplots) in 73 days; ±: standard deviation between replications.

confidence interval: 0.160–270 g) per mole of light ($p_{\sum PPFD}$ < 0.01, F 24.4, df 14; $p_{LP*\sum PPFD}$ 0.61, F 0.28, df 12; n 17) (Fig. 1a). Since the yield response to light was linear, lower light levels during the vegetative stage reduced the final flower yield. In addition, plants in Dynamic LP yielded about 37 g more flowers per square meter than plants in Static LP (p_{LP} 0.05). This resulted in flower increments of 10.4%, 8.9%, and 7.7% for plants in the three Dynamic light treatments 150/700 (2109 mol m^{-2}), 300/700 (2385 mol m^{-2}), and 500/700 (2694 mol m^{-2})

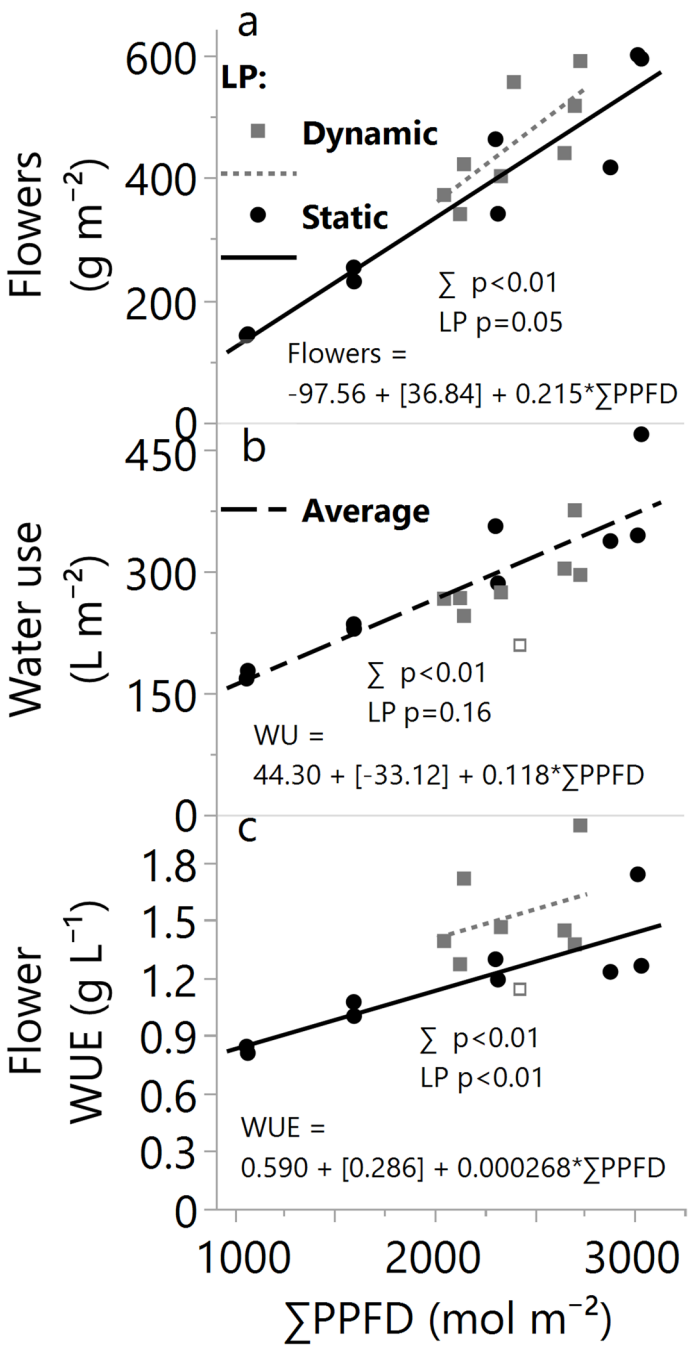


Fig. 1. (**a**) Mass of flowers at 10.2 ± 0.5% moisture content, (**b**) crop water use or WU, and (**c**) flower water use efficiency or WUE (Flowers/WU) based on greenhouse supplemental and solar cumulative photosynthetic photon flux densities (∑ or ∑PPFD) and two lighting programs (LP). The two LP were (1) Static (dots and solid lines): supplemental PPFD of 150, 300, 500, or 700 µmol m^{-2} s^{-1} for 72 days, and (2) Dynamic (squares and dashed lines): supplemental PPFD of 150, 300, or 500 µmol m^{-2} s^{-1} for 28 days, followed by 700 µmol m^{-2} s^{-1} for the final 44 days. In each panel, the equation and p-values represent the regression model equation and its p-values for the main factors (∑ and LP). In the equations, [value] represents the Dynamic LP's effect compared to the Static LP's result if LP $p \leq 0.05$; for average regression outputs, use [value/2]. Empty squares represent a flower mass outlier not included in the analyses. Dots and squares represent the mean per repetition.

compared to plants in the Static treatments at equivalent ∑PPFD (Fig. 1a). Consequently, light enhanced yield as hypothesized, but light removal before the flower growth period also reduced the final flower yield.

The overall increase in flower production could be attributed to numerous factors, such as increased photosynthesis, more light capture due to faster canopy closure and larger to optimal leaf area indexes (LAIs) during production (Supplementary Material Fig. S1), higher remobilizable carbohydrate storage, increased number of inflorescences per plant due to more branches and nodes[7], and a higher number of flowers per inflorescence, as seen in other species[28–30]. The plants likely benefited differently from the combination of the mentioned factors, including light levels and the program (LP).

Effects of light before flower growth on flower yield Similar to the present results, other research on cannabis reported positive impacts on flowers when the lighting was increased during the flowering stage[9,10,26]. Still, this study is the first to analyze and show positive carried-over effects on flower yield from increased lighting before the growth of any flower. More specifically, increasing the lighting levels for 28 days before the flower growth period also increased the flower yield, as represented by the Dynamic-treatment regression line in Fig. 1a. Based on our previous report[7], enhanced flower yield may have been achieved by more production sites (nodes m^{-2}), more resources per node and inflorescence, or both. For instance, we reported a production ratio of 0.41 primary and secondary branches per mole of light during the vegetative stage[7], which should also correlate with more nodes. In the same study, light increased leaf photosynthesis, plant biomass, and LAI, likely increasing light capture and temporary storage of carbohydrates. Furthermore, studies on annual crops have shown that remobilizable carbohydrates (carbohydrates potentially available for flower development and growth) are between 10% and 50% of the stem weight at flowering[31]; meanwhile, cannabis stem mass increased 3 times due to more lighting during the vegetative stage[7]. Therefore, different light effects during the vegetative stage of the crop may have contributed to the increased flower yields in this report.

Effects of light during flower growth on flower yield Besides the positive impact of lighting primarily during the vegetative stage, supplementing with more light in the final 44 days (the flower growth period) also enhanced flower mass (Fig. 1a). Specifically, flower yield increased with higher PPFD during the flowering period under two scenarios: (1) Increased cumulative PPFD (∑PPFD): For example, flower mass increased by 261 g m^{-2} when the treatment changed from 150/150 to 150/700 PPFD (vegetative/flowering), corresponding to an increase in ∑PPFD from 1065 ± 4 to 2109 ± 53 mol m^{-2}. (2) Same cumulative PPFD (∑PPFD): As previously noted, flower mass increased by 37 g m^{-2} under Dynamic lighting (e.g., 300/700) compared to Static lighting (e.g., 500/500), despite similar ∑PPFD values of 2385 ± 47 and 2312 ± 8 mol m^{-2}, respectively.

In both cases, whether ∑PPFD increased or remained similar, applying higher PPFD during the flowering phase relative to the vegetative phase resulted in greater flower yield.

Yield increments observed in the low to high PPFD levels during flowering (i.e., 150/150 to 150/700) were expected based on the relatively low greenhouse solar light levels during the winter experiment (solar DLI at the treatments' canopies: 7.5 ± 0.3 mol m^{-2} day^{-1}). However, comparing the flower growth of plants in Dynamic versus Static treatments under similar cumulative lighting levels is more challenging, as detailed below.

Effects of dynamic vs. static lighting on flower yield Compared to Static Light levels, Dynamic treatments increased flower production under similar cumulative lighting (e.g., Dynamic 300/700 µmol m^{-2} s^{-1} at 2385 mol m^{-2} vs. Static 500/500 µmol m^{-2} s^{-1} at 2312 mol m^{-2}). The boosted flower yield in the Dynamic treatment plants compared to the Static treatment plants was most likely due to increased photosynthesis. The increased growth of the flowers occurred even under lower LAIs, fewer branches, and smaller plant biomass during the first 21–28 days[7] (Supplementary Material Fig. S1). This conclusion assumes that LAI and other previously discussed factors were less decisive than the photosynthetic benefits during flowering under the conditions of this study, such as a minimum LAI of 2 (~ 80% light interception[7] and its rapid expansion at the beginning of flowering (Fig. S1). Moreover, enhanced lighting during the flowering stage promoted a substantial biomass and LAI production (Supplementary Material Figs. S1–S3). Still, the Dynamic LP produced more flowers (Fig. 1a) with the same shoot growth (Fig. S4) compared to the Static LP, implying that the flower production light use efficiency was higher for the Dynamic treatment's plants.

In summary, flower production was likely determined by different light-dependent plant factors along the vegetative and flowering stages. Still, additional light at any point was the primary determinant for crop growth (Fig. 2 and Supplementary Material Fig. S4). These findings highlight the importance of increasing light during the vegetative and flowering cycles while stressing the impact of maintaining or enhancing light levels during flowering.

Crop WU and flower production WUE

Quantifying water use in crop production is crucial for assessing crop feasibility, sizing water infrastructure, reducing the crop water footprint, and ensuring sustainability. However, data on this topic are scarce and exhibit unexplained high variability in cannabis production[5,6]. Crop water use (WU) and efficiency (WUE) can vary depending on the harvested product, production length, and additional environmental factors affecting evapotranspiration beyond light (Supplementary Material Table S1). We previously reported on the effects of light on vegetative growth WU and WUE for the first 21 days[7]. Here, we present data on flower production WU and WUE of *C. sativa* 'Suver Haze' accumulated until day 73.

In this study, light supplementation linearly increased crop evapotranspiration (WU) at a rate of 0.118 L mol^{-1} ($p_{\sum PPFD} < 0.01$), with no differences by the light program (p_{LP} 0.16, $p_{LP*\sum PPFD}$ 0.77) (Fig. 1b). The estimated WU (and 95% confidence intervals) at cumulative light levels of 1065 and 2978 mol m^{-2} were 153 (86–212) and 379 (320–433) L m^{-2}, respectively (Fig. 1b). On the other hand, water requirements per gram of flower (1/



Fig. 2. *Cannabis sativa* 'Suver Haze' grown in a greenhouse under different supplemental LED lighting levels and two lighting programs (LP). The two LP were (1) Static (top plants), PPFD: 150, 300, 500, or 700 µmol m^{-2} s^{-1} for 72 days, and (2) Dynamic (bottom plants), PPFD: 150, 300, or 500 µmol m^{-2} s^{-1} for 28 days, followed by 700 µmol m^{-2} s^{-1} for the final 44 days. Numbers between [] indicate cumulative solar plus supplemental PPFD from planting to harvest (73 days) in moles per square meter. Arrows: 51 cm.

flower WUE) decreased with additional lighting in both LP treatments, resulting in a linear increase in flower WUE ($p_{\sum PPFD} < 0.01$, $p_{LP*\sum PPFD}$ 0.21), from 1.02 to 1.53 g L^{-1} (Fig. 1c). More specifically, a 2.8-fold light increase in the Static LP (1065 to 2978 mol m^{-2}) led to a 4.1-fold growth in flower yield, a 59% flower WUE increase (0.86 to 1.39 g L^{-1}), and a 37% water footprint reduction (1.14 to 0.72 L g^{-1}). Additionally, the Dynamic LP increased WUE by 0.286 g more than Static LP for all the light treatments ($p_{LP} < 0.01$, $p_{LP*\sum PPFD}$ 0.21); this resulted in 24.8%, 23.3%, and 20.6% increases in flower WUE when comparing Dynamic and Static treatment model equations at 2109, 2385, and 2694 mol m^{-2}, respectively (Fig. 1c). Consequently, the first hypothesis, that increasing supplemental lighting would improve flower mass yield and water use efficiency, was supported by treatments with increased lighting throughout the entire production cycle (Static treatments) and during the vegetative stage (Dynamic treatments). In contrast, the second hypothesis—that reducing supplemental light levels before the flowering period would have minimal impact on final flower yield and would increase water use efficiency—was not supported, as reduced lighting negatively affected flower production, resulting in lower yield and WUE.

The flowering results followed the same patterns observed during the vegetative growth stage. The increase of supplemental light linearly increased plant biomass and branch production on day 21[7] and flower production and plant growth on day 73 (Figs. 1a and 2, and Supplementary Material Fig. S4) while improving WUEs. Furthermore, the increments in flower production WUE with higher ∑PPFD (Fig. 1c) were likely due to relatively higher water usage at lower light levels, as previously described[7]. In addition to increased WUE with the increase of ∑PPFD, the Dynamic LP also improved WUE by enhancing flower biomass (Fig. 1a). Therefore, we conclude that the increased flower production WUE (flower yield per plant or m^{-2} divided by the evapotranspirated water per plant or m^{-2}) was primarily caused by the enhanced flower yield from increased Dynamic or Static cumulative light levels.

Leaf net photosynthesis (*A*), transpiration (*E*), stomatal conductance (g_{sw}), and water use efficiency (WUE)

The gas exchange information from top leaves, which captures most of the incoming light and can contribute to other leaf and organ photoassimilates, provides critical information about crop biomass gain and water use. For instance, we previously demonstrated strong positive effects of instantaneous radiation (PPFD) on leaf photosynthesis, transpiration, stomatal conductance, and water use efficiency (leaf WUE = *A*/*E*), as well as some effects of cumulative light at the end of the vegetative stage[7]. However, in addition to light, ontogeny and other factors can modify leaf morphology and physiology, including photosynthesis[10,32–34]. Therefore, leaf gas exchange measurements were taken at the end of the flowering stage. The analysis included *A*, *E*, g_{sw}, and WUE of light-exposed leaves grown under four cumulative light levels (∑PPFD), ranging from 1065 to 2978 mol m^{-2} or 14.7 to 41.1 mol m^{-2} d^{-1} (Table 1). The leaf measurements were performed under seven PPFD levels, ranging from 150 to 1600 µmol m^{-2} s^{-1} (Fig. 3a–c). The objective was to compare the responses of leaves developed under different lighting environments.

Photosynthesis

Higher PPFD levels increased leaf net photosynthesis in every evaluated leaf across treatments (Fig. 3a). The total cumulative lighting (the ∑PPFD treatments) also influenced *A*. For instance, the net photosynthetic saturation (A_{max}) was estimated at 26.2 and 29.4 µmol of CO_2 m^{-2} s^{-1} for leaves grown under the lowest and

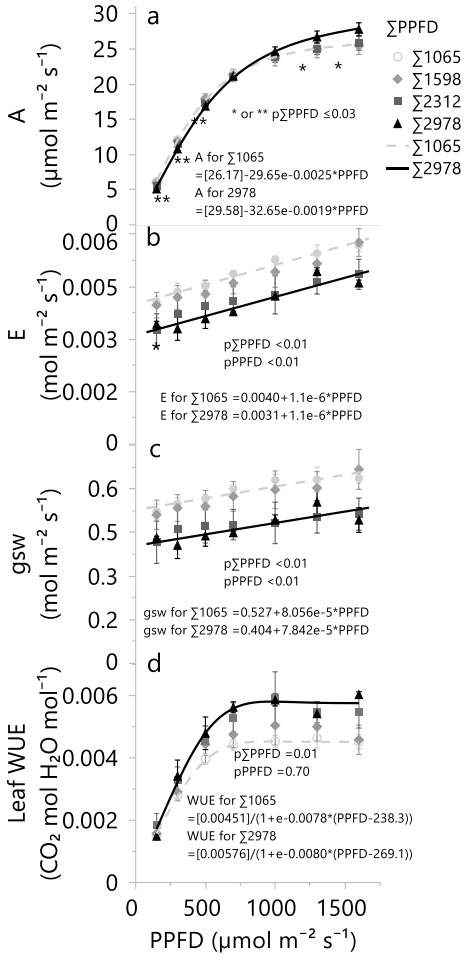


Fig. 3. Net photosynthesis (**a**), transpiration (**b**), stomatal conductance (**c**), and water use efficiency (**d**) of top leaves under instantaneous PPFD levels from 150 to 1600 µmol m^{-2} s^{-1}. Expanded and fully light-exposed leaves were developed under greenhouse crop conditions of four cumulative light treatments (∑PPFD). For clarity, equation lines are shown only for the lowest and highest ∑PPFD (1065 and 2978 mol m^{-2}). In panel (a), the asterisks represent the ∑PPFD effect at each PPFD level using one (*) and two (**) tailed t-tests from linear regression analysis. The panels (b) to (d) were analyzed using a split-plot design and all PPFDs for *E* and g_{sw} or PPFDs ≥ 700 µmol m^{-2} s^{-1} for Leaf WUE; the mixed models assumed equal treatment slopes based on the interaction significances ($p_{\sum PPFD*PPFD}$ 0.99). [] represent maximum *A* and WUE in panels (a) and (d), respectively.

highest ∑PPFDs, respectively. Compared to the lowest ∑PPFD treatment, the leaves grown under the greatest ∑PPFD had an enhanced *A* of 7% to 9% at PPFDs of 1300 to 1600 µmol m^{-2} s^{-1} but 5% to 17% lower *A* at measured PPFDs of 500 to 150 µmol m^{-2} s^{-1} ($p_{\sum PPFD} \leq 0.03$; Fig. 3a). The PPFD and the supplemental light treatment (∑PPFD) effects on *A* may be explained firstly by an increased photosynthetic capacity due to thicker leaves[31], more Rubisco[35], and more chlorophyll[36], and secondly, by enhanced respiration rates driven by higher energy requirements under high-light acclimated leaves[35,37]. Almost identical responses on *A* from increased PPFD and ∑PPFD treatments were observed during the vegetative stage[7]. Furthermore, similar trends and A_{max} values were reported during the flowering of a different cultivar and growing conditions[10].

Concerning the effect of ontogeny on photosynthesis, our previous and present studies point out a potential but expected reduction in A_{max}[38]. For instance, from the vegetative to the flowering stages, A_{max} dropped from 33.7 to 39.5 to 26.2–29.4 µmol of CO_2 m^{-2} s^{-1} in leaves grown at daily light integrals of 17.9–51.8 and 14.7–41.1 mol m^{-2} d^{-1}, respectively. Deterioration of A_{max} from vegetative to flowering or during the flowering stage has been reported in different cannabis cultivars and growing conditions[10,39]. Despite the normal deterioration of A_{max}, the values reported in 'Suver Haze' and other cannabis cultivars during flowering seem to be among the highest A_{max} among C3 crops based on comparable ontogeny and CO_2 levels[40–44]. The smaller reduction in A_{max} in the present study compared to a previous report on cannabis[38] may be explained by maintaining the leaf under optimal conditions for maximum photosynthesis, such as high irradiance, proper spectrum, and adequate irrigation and nutrition, among others[40,42,45], which may have enabled a higher production and maintenance of chlorophyll, stomatal, and other leaf functions for an extended period.

In conclusion, the increased photosynthetic rates in cannabis leaves and their acclimation to higher light levels were positively associated with greater flower yield (Fig. 1a) and general crop growth (Fig. 2; Supplementary Fig. S4). However, the enhanced growth resulted mainly from higher instantaneous light levels (LED plus solar), rather than from morphological or physiological differences between leaves observed in this study.

Transpiration and stomatal conductance

As leaf photosynthesis increased at progressively lower rates with more PPFD (Fig. 3a), transpiration (Fig. 3b) and stomatal conductance (Fig. 3c) exhibited a linear increase ($p_{\sum PPFD} < 0.01$). Moreover, both ratios, g_{sw} to PPFD and the E to PPFD, decreased with light (PPFD). For instance, the g_{sw}/PPFD ratios at 150 and 1600 (PPFD) were respectively 3141 ± 476 and 355 ± 58 mol mol^{-1}, an 8.9 fold difference, while E/PPFD ratios were 24.7 ± 3.7 and 3.2 ± 0.4 mol mol^{-1}, respectively, a 7.7 fold difference. In other words, water and CO_2 conductance would have been considerably high at night with no light or increased drastically with low PPFDs in the morning. We previously reviewed consistent results during the vegetative stage of this crop and similar responses across species, explaining the negative relationship between low light levels and water use efficiency[7].

While an increase in instantaneous PPFD enhanced leaf transpiration and stomatal conductance, leaves grown under increased cumulative PPFDs (∑PPFD) exhibited reductions in transpiration and stomatal conductance rates at the same PPFD ($p_{\sum PPFD} < 0.01$, $p_{PPFD*\sum PPFD}$ 0.99) (Fig. 3b and c). For instance, *E* was reduced by 0.94 ± 0.22 mmol m^{-2} s^{-1} from low to high light-grown leaves when comparing at the same PPFD (Fig. 3b). This *E* reduction resulted in water savings of up to 22.5% when measuring leaves under a low PPFD (150 µmol m^{-2} s^{-1}). Therefore, while increased instantaneous lighting (PPFD) promoted *E* and g_{sw}, prolonged exposure to more light (∑PPFD) reduced both *E* and g_{sw}.

Stomatal regulation

Stomatal conductance is the main factor regulating transpiration and photosynthesis and depends on several factors. The reduction of up to 22.5% in transpiration from low to high light-grown leaves could have resulted from potential changes in stomatal density, size, and/or biochemical regulation affecting stomatal aperture, and consequently, conductance (g_{sw}). For instance, leaves developed under more lighting are likely to increase stomatal density[35,46–50], while the size of the stomata may be reduced. In other words, a negative correlation can exist between stomatal density and size[51,52]. Still, smaller stomata are associated with faster stomatal responses and water savings[51]. Aperture restrictions on stomata could also be explained by changes related to abscisic acid (ABA), the primary hormone regulating stomata. Robust ABA signaling has been correlated with increased crop WUE and water-savings[44]. More specifically, applications of ABA increased the minimum temperature to open stomata[53] and reduced stomatal aperture under increased lighting[54,55]. Moreover, positive correlations of ABA with increased lighting, light stress-suppressing metabolites, and other stressors are reported[56,57]. Consequently, we hypothesize that changes in ABA requirements[58] or concentrations due to prolonged high-lighting exposure or ∑PPFD could explain the transpiration reduction results in the present study.

Regarding photosynthesis and stomatal aperture, potential reductions in photosynthesis could neither be confirmed nor ruled out based on the observed declines in g_{sw} in this study (Fig. 3a)[59]. This ambiguity likely arises because photosynthetic limitation depends not only on stomatal aperture but also on ambient CO_2 concentration and the combined resistances across all leaf components. In conclusion, understanding the effects of light on stomata is crucial for maximizing photosynthesis and minimizing water loss through transpiration.

Leaf WUE

Leaf water use efficiency (*A*/*E*) varied based on *A* and *E* responses to increasing PPFD (Figs. 3a and b). Leaf WUE increased with more PPFD until reaching estimated maximum values of 0.0045 and 0.0058 mol CO_2 mol H_2O^{-1} for crops grown under the lowest and highest cumulative light levels, respectively (Fig. 3d). Furthermore, the WUEs at 700 µmol m^{-2} s^{-1} (PPFD) were already at 97% of the maximum WUE values, with non-statistical differences in WUE from 700 to 1600 µmol m^{-2} s^{-1} of PPFD (Fig. 3d). Consequently, leaf WUE at and beyond 700 µmol m^{-2} s^{-1} increased solely with the long-term impact of growing under increased cumulative lighting ($p_{\sum PPFD}$ 0.01), with WUE of 0.0045 ± 0.0002 and 0.0057 ± 0.0003 mol CO_2 mol H_2O^{-1} at the lowest and highest

∑PPFD treatments, respectively. The difference between the two means represented a 27% increase in net CO_2 fixation per mole of water and a 21% water saving per mole of CO_2 when comparing top leaves of plants grown under the highest light intensity treatment to those under the lowest ∑PPFD. However, leaf WUE decreased below 700 µmol m^{-2} s^{-1} PPFD, reaching a minimum of 0.0016 ± 0.0003 mol mol^{-1} at 150 µmol m^{-2} s^{-1} PPFD across all ∑PPFD treatments. Consequently, the increment in leaf WUE between 150 and 700 PPFD was 0.0029 ± 0.0000 for the low ∑PPFD treatment and 0.0041 ± 0.0002 for the high ∑PPFD, respectively, with the two being significantly different ($p < 0.01$). Consequently, PPFD increments were crucial for immediately reducing the water footprint of physiological processes (mol H_2O mol CO_2^{-1}), while the long-term light acclimation changes allowed for some additional photosynthetic capability and considerably reduced water usage.

Our general hypothesis that long light exposure enhancement (∑PPFD) would increase leaf efficiency was correct for leaf water use (*E*) under all PPFD conditions (150–1600 µmol m^{-2} s^{-1}), but net carbon assimilation (*A*) and WUE (*A*/*E*) depended on the instantaneous lighting levels or PPFD.

In summary, increased ∑PPFD showed significant improvements (reductions) in *E* at all PPFD levels, while increased PPFD dominated increments in carbon fixation at most PPFD levels (150–1000 µmol m^{-2} s^{-1}), being also a key factor for water savings from 150 to 700 µmol m^{-2} s^{-1}. Additional benefits were achieved in flower growth and water savings by increasing the PPFD during the flowering growth stage in the form of Dynamic PPFD treatments. Specific values from the leaf responses are likely to be skewed toward higher PPFD levels for a canopy. For instance, canopies, especially at higher LAIs, are expected to respond more linearly and require higher light levels to achieve a plateau growth response compared to a single leaf or small LAI response to light[60,61], as seen in the present study's crop growth linear responses and LAI conditions (Fig. 1a, Supplementary Material Figs. S1 and S4). The leaf and crop responses in the present study highlight the strong positive correlation between photosynthesis, flower production, and water savings and offer valuable insights for crop production optimization.

Light effects on flower partitioning and quality

Flower partitioning

Research reports have shown that lighting augmentation increases cannabis crop flower production and sustains or increases the cannabinoid and terpene concentrations in top flowers[10–12,26,27]. However, the concentration, particularly of terpenes, typically drops from apical to non-apical flowers and from upper to lower branches or nodes[11,21–23]. Therefore, determining where the increased flower yield occurs within the plant is crucial for understanding yield and quality. Moreover, knowing where flowers are formed based on resources can be essential for making decisions about plant trimming[21], size, density, and inter-canopy lighting[62]. Consequently, this study separated the flower biomass based on the main stem's top, middle, and bottom nodes. The relative flower yield distribution along the main stem was significantly altered ($p_{Static\ \sum PPFD} < 0.01$) by an increase in Static light levels from 1065 to 2978 mol m^{-2}, spanning production DLI levels between ~ 15 and 41 mol m^{-2} d^{-1}. Specifically, the flower yield decreased at the top nodes from 41.5% to 19.1% and at the middle nodes from 43.8 to 34.1%, while it increased substantially at the bottom nodes from 14.7 to 46.8% (Fig. 4a) when Static light levels increased from 1065 to 2978 mol m^{-2}, respectively. A similar pattern was observed in another species[28]. In comparison, the results from the Dynamic treatments showed high variability but exhibited a similar trend at the top nodes ($p_{Dynamic\ \sum PPFD} = 0.04$; Fig. 4b). This variability, along with the observed results, can be attributed to light interception and the light supplementation provided prior to flower development, rather than the light supplementation during the flower growth period, which remained consistent across treatments (Table 1).

The effects of increased lighting in the static treatments resulted in increased flower growth from the top to the bottom nodes of the plants (Fig. 4a). Moreover, the distribution of flower yield indicates that resources for flower and metabolite production were more limited at the lower branches compared to the upper nodes and branches. Several factors may have contributed to the enhanced flower production on lower branches, including increased photosynthesis due to increasing light levels from 1065 to 2978 mol m^{-2}, enhanced translocation of photoassimilates from upper to lower branches, and greater carbohydrate reserves in larger plants grown under higher-light conditions. Conversely, under the lowest lighting conditions (∑PPFD of 1065 mol m^{-2}), the upper 14 nodes (approximately two-thirds of the main stem nodes from the first branch) produced 85% of the total yield. Therefore, under lower light conditions (∑PPFD of 1065 mol m^{-2}), limiting the number of nodes to 14 from the first branch may be a more effective strategy to reduce the duration of the vegetative stage, shorten the time to harvest, and avoid producing flowers with lower cannabinoid and terpene concentrations. This strategy could be most efficiently implemented by reducing the vegetative period to less than 23 days and increasing plant density to achieve similar LAIs as in Fig. S1 for low-light conditions (~ 15 mol m^{-2} d^{-1}). Still, the final number of branches and nodes may also vary with the cultivar[63], target flower size (inverse to the number of branches at the same DLI), and pinching or topping procedures[64].

In conclusion, lighting augmentation can significantly alter flower partitioning along the cannabis's stem. Higher Static light levels shifted flower production from the upper to the lower nodes, likely due to enhanced photosynthesis and resource translocation, thereby improving overall yield. Although terpene concentrations typically decline from apical to non-apical flowers, strategic adjustments in node management and trimming could optimize both yield and quality. Future studies should also consider cultivar differences and the effects of CO_2 enrichment.

Flower terpenes and cannabinoids

For upper flowers, where the light is primarily intercepted and resources are likely prioritized[7,65], the literature reports terpene and cannabinoid concentrations that remain stable or increase in terpenes in response to higher

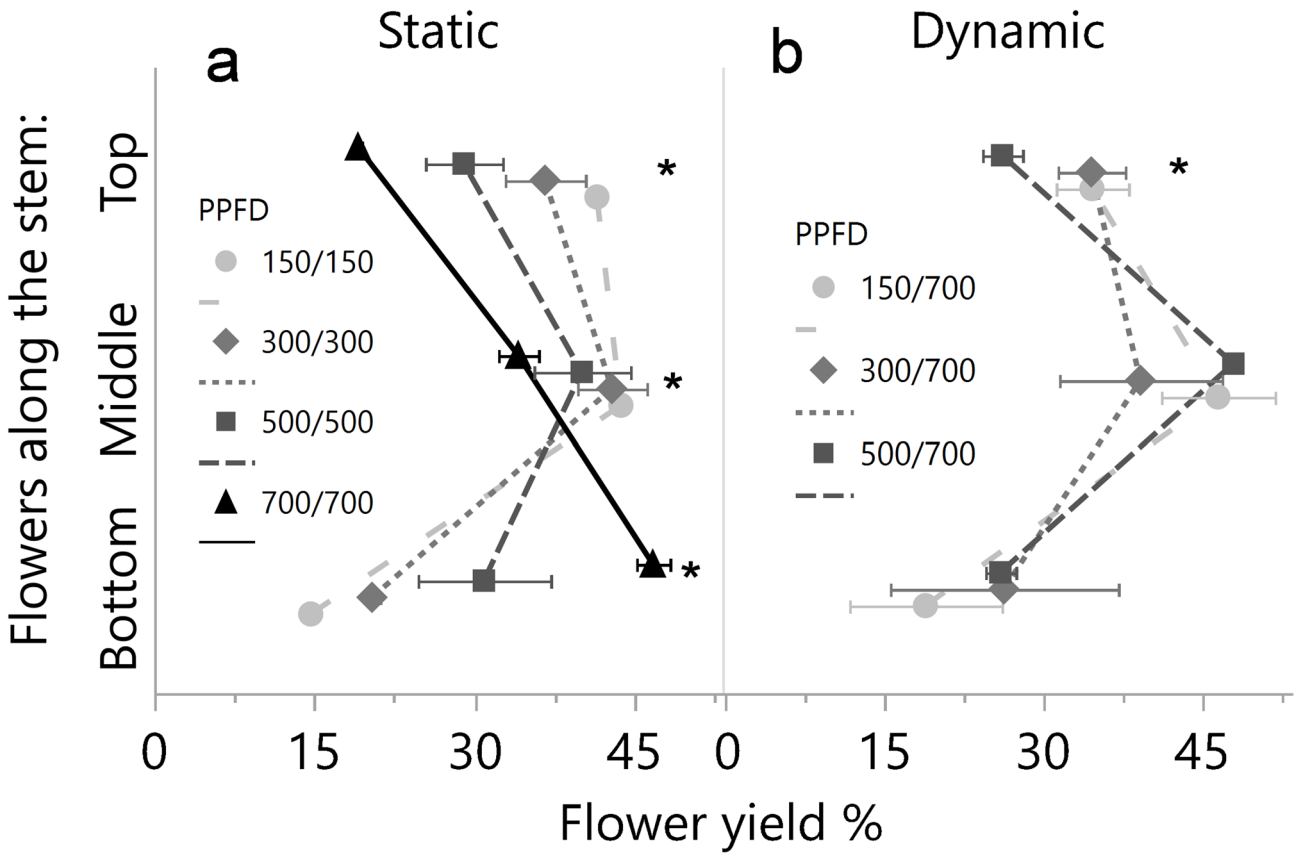


Fig. 4. Effects of the greenhouse lighting level and two lighting programs (LP) on the flower yield relative contribution between the main stem's top, middle, and bottom nodes. PPFD indicates the supplemental light levels before and during flower growth. The two LP were **a**) Static, PPFD: 150, 300, 500, or 700 µmol m^{-2} s^{-1} for 72 days, and **b**) Dynamic, PPFD: 150, 300, or 500 µmol m^{-2} s^{-1} for 28 days, followed by 700 µmol m^{-2} s^{-1} for the final 44 days. Asterisks: regression analysis p-values ≤ 0.04 for ∑PPFD by flower's origin and LP. Plot symbols: the treatment average ± standard error (n 2 to 3).

light levels[10,12,26,27]. The effect of lighting on secondary metabolites from flowers on lower branches has received less attention[11]. However, the concentration of these metabolites frequently drops from the top to the bottom of the plant[21–23]. This study evaluated the impact of the light on terpene and cannabinoid concentrations and compositions in flowers from the fourth-most apical node to the bottom of the plant. The evaluated flowers represented 92.5 ± 2.4% of the total, with limitations in photoassimilate availability expected to be significant compared to upper flowers.

Terpenes

The terms terpenes and terpenoids are used interchangeably in the literature[66] but also refer to non-oxygenated and oxygenated terpenes, respectively[67]. In this study, terpenes refer to oxygenated and non-oxygenated terpenes in the mono and sesquiterpene/terpenoid groups[12,25].

Figure 5a results indicate that the increase in light (∑PPFD) linearly decreased terpene concentration ($p_{\sum PPFD}$ = 0.01, p_{LP} = 0.03; $p_{\sum PPFD*LP}$ =0.08). However, this response was statistically significant only under the Dynamic LP treatment, 31.4e^{-5}% mol^{-1} m^2 (p 0.03), while the Static LP treatment showed a non-significant slope, -6.1e^{-5}% mol^{-1} m^2 (p 0.13). The results in the Dynamic LP treatments with the same light levels during the flower growth period contradict previous research findings, which suggest that plants exposed to higher light conditions (during flowering) should also exhibit stable to increased terpene levels[10–12,26]. Consistent with prior literature, this study found that terpene concentration increased with higher light intensity during flowering and lower light during the vegetative phase. Specifically, terpene levels were significantly higher under the 150/700 treatment (∑PPFD: 2109) compared to the 150/150 treatment (∑PPFD: 1065), with a one-sided t-test indicating significance (p 0.026). The reduction in terpene concentration under the Dynamic LP treatments occurred despite identical light levels during flowering and increased light exposure before flowering. This finding contradicts the expected physiological responses reported under similar growing conditions, such as enhanced stomatal development in future leaves triggered by increased light on existing foliage[68]. A more plausible explanation is that increased shading and higher photoassimilate demand, resulting from a greater leaf area index and total plant biomass due to elevated light levels during the vegetative stage (Fig. 2; Supplementary Figs. S1, S2, S4), limited terpene accumulation in flowers on the lower branches. In contrast, applying more light during the flowering stage to

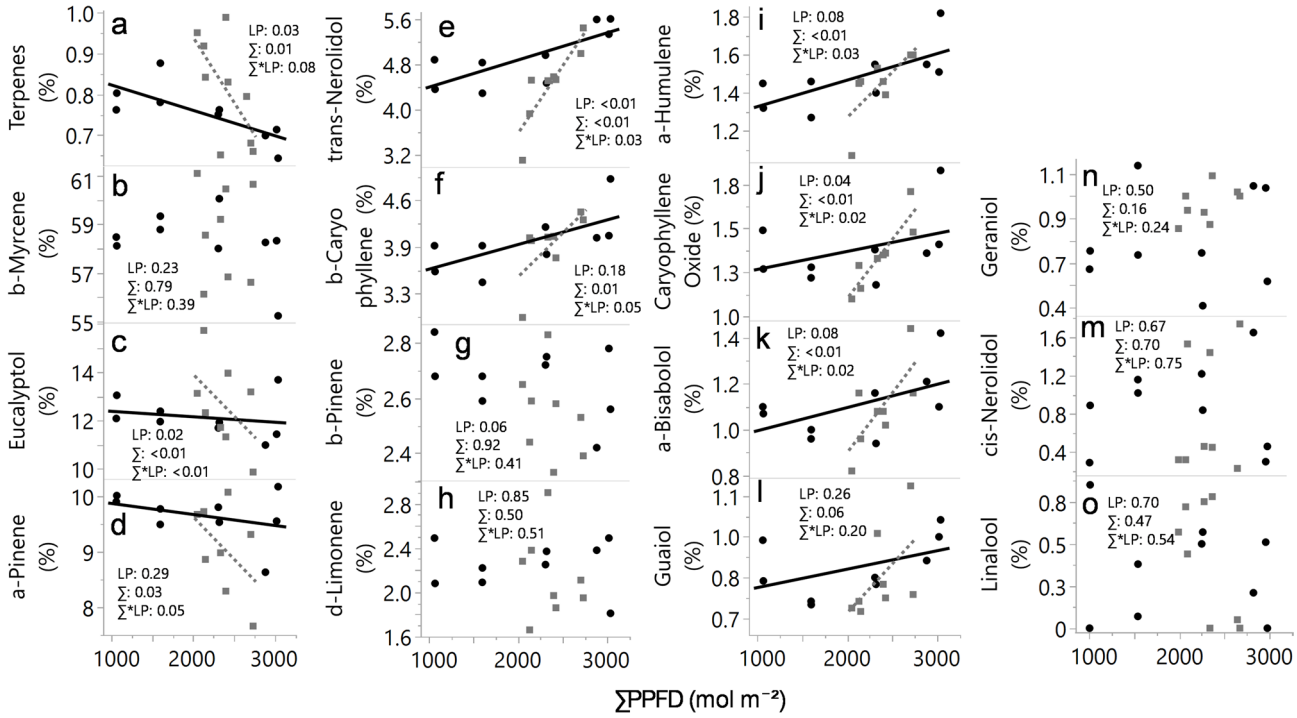


Fig. 5. Effects of the cumulative solar plus supplemental lighting (∑PPFD) and two lighting programs (LP) on **a**) the concentration of 14 terpenes on flower[z] mass (g g^{-1} * 100) and **b**) to **o**) the percentual mass contribution of an individual terpene to the sum (g g^{-1} *100). The two LP were (1) Static (dots and solid lines), PPFD: 150, 300, 500, or 700 µmol m^{-2} s^{-1} for 72 days, and (2) Dynamic (squares and dashed lines), PPFD: 150, 300, or 500 µmol m^{-2} s^{-1} for 28 days, followed by 700 µmol m^{-2} s^{-1} for the final 44 days. ∑ (∑PPFD), LP, and ∑*LP are the linear regression p-values from the main factors and their interaction. Because of the interaction factor, the main factor significances represent differences or no differences at a centered ∑PPFD of ~ 2200 mol m^{-2}. Square and dot marks are the averages per replication. The Y-axes were stretched out to show minor changes. [z]All flowers below the third-most apical node along the main stem.

smaller plants (e.g., treatment 150/700) may have improved overall resource availability, promoted resource partitioning to the lower branches, and enhanced light penetration to the lower canopy area. As a result, the availability of photoassimilates in flowers on the lower branches may have increased, supporting greater terpene production.

Based on the literature and the results of this study, resources were prioritized first for upper-node flowers and later for lower-node flowers (Fig. 4). For these reasons, optimizing terpene concentration should focus on a balanced leaves-to-flower ratio (source-to-sink ratio), crop architecture[21], and lighting levels on and within a canopy, including inter- and sub-canopy lighting[11,62].

In addition to genetic effects, other factors strongly affecting secondary metabolites are flower age and trichome type and maturity, which will be discussed in the following sections. In conclusion, our hypothesis that light enhancement would increase the flower quality (terpene concentration) may hold for apical flowers[10,11,26] or under lower canopy biomass, i.e., 150/150 to 150/700 (more light, less biomass) compared to 150/150 to 700/700 (more light and biomass).

Single terpenes

In this study, 22 cannabis terpenes were analyzed. Still, some were not detected (camphene, α- and γ-terpinene, Δ^3-carene, β-ocimene, p-cymene, terpinolene, and isopulegol), which could be due to genetic and tissue differences[69–71]. On the other hand, the effects of ∑PPFD and LP on seven monoterpenes (β-myrcene, eucalyptol, α-pinene, β-pinene, δ-limonene, geraniol, and linalool) and seven sesquiterpenes (trans-nerolidol, β-caryophyllene, α-humulene, caryophyllene oxide, α-bisabolol, guaiol, and cis-nerolidol) are shown from the highest to lowest individual contribution (Figs. 5b-o) to the sum of terpenes (Fig. 5a). β-Myrcene contributed more than half to the sum of terpenes. Still, there was no evidence of treatment effects, $p_{\sum PPFD,\ LP,\ \sum PPFD*LP} \geq 0.23$ (Fig. 5b). Increased cumulative lighting on Dynamic treatments negatively affected the monoterpenes eucalyptol and α-pinene ($p_{Dynamic\ \sum PPFD} \leq 0.03$). On the other hand, the concentrations did not change on the Static treatment plants ($p_{Static\ \sum PPFD}$ 0.57); in other words, the ∑PPFD effect on those monoterpenes depended on the light program (Figs. 5c and d). In contrast to monoterpenes, several sesquiterpenes (trans-nerolidol, β-caryophyllene, α-humulene, caryophyllene oxide, and α-bisabolol) were positively influenced by cumulative light exposure ($p_{\sum PPFD} \leq 0.01$). However, this response varied depending on the lighting program ($p_{\sum PPFD*LP} \leq 0.05$) (Figs. 5e, f, i, j, and k). Similarly, higher lighting treatments were associated with an increased proportion

of sesquiterpenes relative to total terpenes or a reduction in monoterpene levels, with the specific outcome likely modulated by the lighting program (Supplemental Material Fig. S5a, b, and c).

In cannabis, relatively lower contents of monoterpenes than sesquiterpenes have been reported in younger versus mature flowers[24,25] and can be explained based on the trichome type and maturity[20]. Additionally, the concentration of some of the main monoterpenes and total terpene percentage in apical flowers decreased with reduced top lighting[10]. Similarly, another study found that the main monoterpenes in the upper flowers were reduced by restricted sub-canopy lighting (lighting from the floor up)[11]. Moreover, cannabis upper inflorescences showed fewer terpenes and even some contrasting changes in specific terpene accumulation when clone plants grew in low-altitude, with reduced solar radiation and increased temperature, compared to high-altitude plants[72]. Thus, the literature evidence suggests that changes in flower/trichome maturity and available resources could explain variations in the terpene concentration and composition shown in Fig. 5a–o.

Cannabinoids and single cannabinoids

This study demonstrated that lower-node flowers can contribute considerably to the final yield, especially under increased lighting. However, as discussed, secondary metabolite concentrations can be reduced in lower flowers. This report investigated the effects of lighting augmentation on the main cannabinoids in flowers below the main stem's third node. Increasing light from 1065 to 2978 mol m^{-2} (∑PPFD) reduced the main cannabinoids' concentration (CBD + CBDA + THC + THCA + CBGA + CBG + CBC) by 13.0%, and the concentrations of total CBD and total THC decreased by 13.4 and 8.9%, respectively ($p_{\sum PPFD} \leq 0.01$, $p_{LP \text{ and }} p_{\sum PPFD*LP} \geq 0.46$) (Fig. 6a–c). However, the relative contribution of specific cannabinoids to the sum of cannabinoids was not affected statistically by any treatment or their combination ($p_{\sum PPFD, LP, LP*\sum PPFD} \geq 0.32$), as well as the CBD/THC and CBDA/THCA ratios ($p_{\sum PPFD}$ 0.78 and 0.35; p_{LP} 0.07 and 0.17; $p_{LP*\sum PPFD}$ 0.81 and 0.59, respectively). The average cannabinoid relative composition and CBD/THC ratios are available in Supplementary Material Table S3. In conclusion, the results did not support our hypothesis that light enhancement would increase cannabinoid concentration. Ultimately, considering the relatively small changes in cannabinoid concentration per mass in the evaluated flowers (92.5%) and assuming noncritical changes in the remaining flowers (7.5%), cannabinoid production per area would be greatly increased by more lighting ($p_{\sum PPFD} < 0.01$, p_{LP} 0.13, $p_{LP*\sum PPFD}$ 0.63) (Fig. 6d) due to increased flower mass yield (Fig. 1a).

The reduced cannabinoid accumulation observed in Fig. 6a may be due to changes in the type, density, size, and fullness of the average glandular trichome[20,73] based on more flowers at the lower branches (Figs. 1a and 4) and perhaps fewer photoassimilates per flower, as previously discussed. For instance, younger flowers (associated with reduced density, size, fullness, and metabolite production trichomes) showed lower cannabinoid concentrations across chemotypes (high CBD, high THC, and CBD ≈ THC chemotypes)[20,25]. Another potential factor influencing cannabinoid concentration is variability in the contributions of individual cannabinoids[25]. However, reports have shown that for the two predominant cannabinoids, the CBDA/THCA ratios in flowers remained consistent between weeks 3 (21.0 ± 1.4) and 11 (21.6 ± 1.6) across three high-CBD cultivars. Additionally, trichome sampling has revealed minimal cannabinoid variation across different trichome types[20]. These findings align with our observed cannabinoid profiles and a CBDA/THCA ratio of 20.6 ± 0.7 (Supplementary Material Table S3). Consequently, like terpenes, the evidence suggests that changes in flower maturity and perhaps available resources could explain the reduction in cannabinoid concentration without significant effects on composition. Nevertheless, there were some differences between terpenes and cannabinoid synthesis, which are discussed below.

Despite overall similarities in the responses of the evaluated terpenes and cannabinoids, there were differences in concentration and the contribution of individual secondary metabolites to the total sum in response to ∑PPFD and LP treatments (Fig. 5a–o and Fig. 6a–c, and Supplementary Material Table S3). This is consistent with previous results on cannabis grown from low to very high light levels, 120 to 1800 µmol m^{-2} s^{-1} (PPFD), where light increased linearly specific monoterpenes (up by 52–95%; R^2 0.32–0.46) and the terpene concentration (up by 44%; R^2 0.32) but did not affect cannabinoids in apical flowers[10]. However, in another study with reduced differences in lighting (600 vs. 1000 µmol m^{-2} s^{-1} of PPFD), neither terpenes nor cannabinoids varied in the top flowers[26]. All this indicates that light affected terpene and cannabinoid syntheses differently, as previously reported[20]. Several factors could explain the effects observed in the present study. Firstly, the cannabinoid mass constituted a significant part of the flower mass (cannabinoids 18.6 ± 1.6%), especially when compared to terpenes (0.8 ± 0.1%). This likely required significant photoassimilate levels[74]. Therefore, the impact of larger flower biomass in Dynamic treatments should also be considered. More specifically, resources available in smaller plants under the Dynamic LP, such as treatment 150/700, might have been allocated to more flowers and terpenes instead of additional cannabinoids. Secondly, even if the light was intercepted at lower branches, some photoassimilates may still be translocated to upper flowers, as previously reported with apical vegetative growth[7]. Thirdly, terpenes might be related to more critical and dynamic functions, such as solar or other stress conditions, even at low terpene concentrations. For instance, compared to low-altitude-grown cannabis, clone plants grown at high altitude with more lighting and reduced temperature produced more terpenes and specific flavonoids -pigments associated with protection from more lighting and other stressors[75]. On the other hand, cannabinoids were less affected than terpenes by more extreme environmental conditions[72]. Moreover, mild water stress increased some terpenes threefold in a few days, while prolonged and severe stress normalized the terpene values to those of non-stress thyme plants[76]. Minor leaf water stress caused by lower soil moisture and lower leaf water potentials caused by higher radiation[77] might trigger similar terpene responses. However, more research is needed to investigate this hypothesis. In conclusion, multiple factors may have simultaneously affected the synthesis of all and/or specific secondary metabolites in addition to light levels.

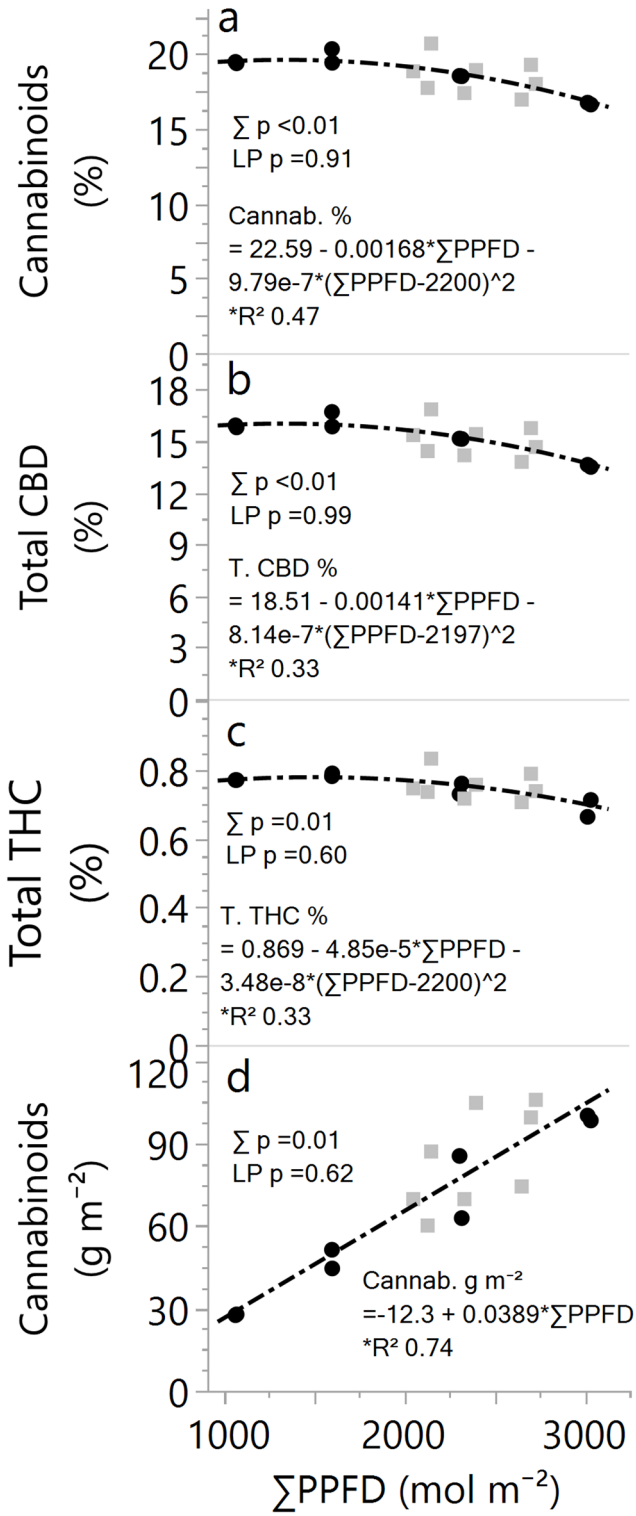


Fig. 6. Effects of the cumulative solar plus supplemental LED lighting (∑PPFD) on the concentration of (**a**) seven cannabinoids, (**b**) total CBD, and (**c**) total THC on flower mass (kg kg^{-1} * 100), as well as the (**d**) cannabinoid production per area from flowers. Two LED light programs (LP) were used: (1) Static (dots), PPFD: 150, 300, 500, or 700 µmol m^{-2} s^{-1} for 72 days, and (2) Dynamic (squares), PPFD: 150, 300, or 500 µmol m^{-2} s^{-1} for 28 days, followed by 700 µmol m^{-2} s^{-1} for the final 44 days. Lines, p-values, and equations from linear regression analyses with significant linear terms for ∑PPFD ($\sum p \leq 0.01$); the quadratic terms ($\sum^2 p \geq 0.55$) strongly increased the adjusted R^2 (*R^2) and yielded homoscedasticity compared to linear term models. The interaction factors (LP*∑PPFD) in the models were not significant ($p \geq 0.49$). Squares and dots are the averages per replication.

As mentioned, optimal plant function under more demanding growing conditions, such as increased lighting, may depend on small changes in secondary metabolites. However, the effects of relatively small variations in metabolites from cannabis products on humans can be challenging to predict. For instance, most cannabis terpenes (α-humulene, geraniol, linalool, and β-pinene) evaluated in mice induced similar cannabinoid tetrad activities, with differences in the size of the effects depending on the terpene type and quantity[17]. Consequently, the quantity of terpenes rather than small changes in the relative contribution of those metabolites seemed more crucial. In contrast, β-caryophyllene demonstrated more contrasting differences compared to the previous terpenes. Moreover, for human medical treatments without cannabinoids, some terpene compositions are preferred over other essential oils. Still, the primary terpenes in the oil may not be the active ingredient for treatments[18]. Furthermore, terpene and cannabinoid interactions can enhance therapeutic effects[17]. Therefore, caution should be exercised when commercializing and using natural products, as variation in secondary metabolites may occur.

Conclusions

Solar plus electrical lighting (1065 to 2978 mol m^{-2} or 14.6 to 40.8 mol m^{-2} d^{-1}) during a 73-day vegetative and flowering cycle significantly and linearly increased flower production (0.215 g mol^{-1}) from 131 to 543 g m^{-2} in *Cannabis sativa* 'Suver Haze'. Additional flower growth (~ 37 g) was achieved by partitioning more light towards the flower growth period and less to the vegetative phase. However, reducing lighting during the vegetative phase had the same detrimental effect on flower production as decreasing lighting during the vegetative plus flowering periods, -0.215 g per reduced mole. The enhanced flower production outweighed the significant impact of increased water use from 170 to 395 L m^{-2}. This was the primary cause of the improved water use efficiency, reducing the water footprint by 37% from 1.14 to 0.72 m^3 kg (evapotranspiration per mass of flower). The present and previous results[7] confirm that supplemental light can be used to increase cannabis production and reduce its water footprint. The linear response observed here and the higher DLI levels reported in the literature[10] indicate that additional supplemental lighting, solar radiation, or both could achieve more significant benefits and water savings. Nevertheless, other factors such as photon wavelengths, temperature, and humidity can influence photosynthesis and evapotranspiration and should be further researched in cannabis.

Furthermore, the present results reinforce our previous findings that supplemental lighting can reduce the water footprint of most crops while increasing production[7]. Considering that *Cannabis sativa* cv Suver Haze is an average-performing crop based on flower yield under different environments and among several cultivars[78], we expect that genetic variability in *Cannabis* spp may primarily affect the magnitude of responses compared to other genotypes. In conclusion, this research offers valuable insights into crop physiology and management and provides a foundation for future research aiming for higher light and water use efficiencies.

Materials and methods

Timeline and cultural production practices

The cannabis study was conducted for 73 days (October 9 to December 21, 2020) in a climate-controlled glass greenhouse in Raleigh, North Carolina, USA, using 24-day-old clone plants of *Cannabis sativa* cv Suver Haze (© Oregon CBD, Corvallis, OR). Plants were previously asexually propagated by a commercial greenhouse nursery under a misting system and a photoperiod of 18 h. On October 9, the 24-day-old plug plants were planted in the research greenhouse with a photoperiod of 18 h and an LED PPFD of 90 μmol m^{-2} s^{-1} (day 0). Lighting treatments started on day 1, with a vegetative harvest on day 21 [7]. On day 23, the LED photoperiod, including solar, was adjusted from 18 to 12 h, or 5:00–23:00 to 6:00–18:00 h. On day 28, two plants per treatment were harvested for biomass and LAI analyses and plant density adjustment. On day 29, the plants in each plot and light levels were rearranged to accommodate the increase in lighting for Objective 2. On day 36, the main stem tip was cut off to maintain a uniform canopy height and light levels. On day 73, flower and other final measurements were taken.

Treatments

Table 1 describes the values of the supplemental lighting treatments, which consisted of a combination of supplemental LED and solar radiation. The cumulative solar + LED light level (∑PPFD: cumulative photosynthetic photon flux density) was used to estimate crop responses in 73 days of production. In addition, the Daily Light Integral (DLI = ∑PPFD / 73 days), a widely used indicator, was used to compare and discuss the literature. The treatment differences were caused by a combination of supplemental PPFD levels and two lighting programs (LP). For instance, a static LP maintained LED lighting levels of 150, 300, 500, or 700 μmol m^{-2} s^{-1} from day 1 to 73. In contrast, a dynamic LP maintained LED light levels of 150, 300, or 500 from days 1 to 29 while rising to and maintaining 700 μmol m^{-2} s^{-1} until day 73 (Fig. 7). For light adjustments, spectrum, and equipment, refer to Collado et al.[7].

Greenhouse setup and experimental units

The greenhouse was retrofitted with 12 independent production areas (plots). Each plot of about 1.2 m x 1.2 m had fully automated lighting and irrigation control. From day 1 to 29, there were three plots for each supplemental LED light level of 150, 300, 500, and 700 μmol m^{-2} s^{-1}. From day 29 to 73, there were two plots (n = 2) for each light level of 150, 300, and 500 μmol m^{-2} s^{-1} to fit the 150/150, 300/300, and 500/500 treatments, respectively. The six remaining plots were used at 700 μmol m^{-2} s^{-1} to fit the treatments 150/700, 300/700, 500/700, and 700/700; each 700 PPFD plot grew two treatments, e.g., 150/700 and 300/700; consequently, treatments in the 700 PPFD plots were replicated three times (n = 3 = 6 plots * 2 treatments per plot / 4 treatments). There was no physical

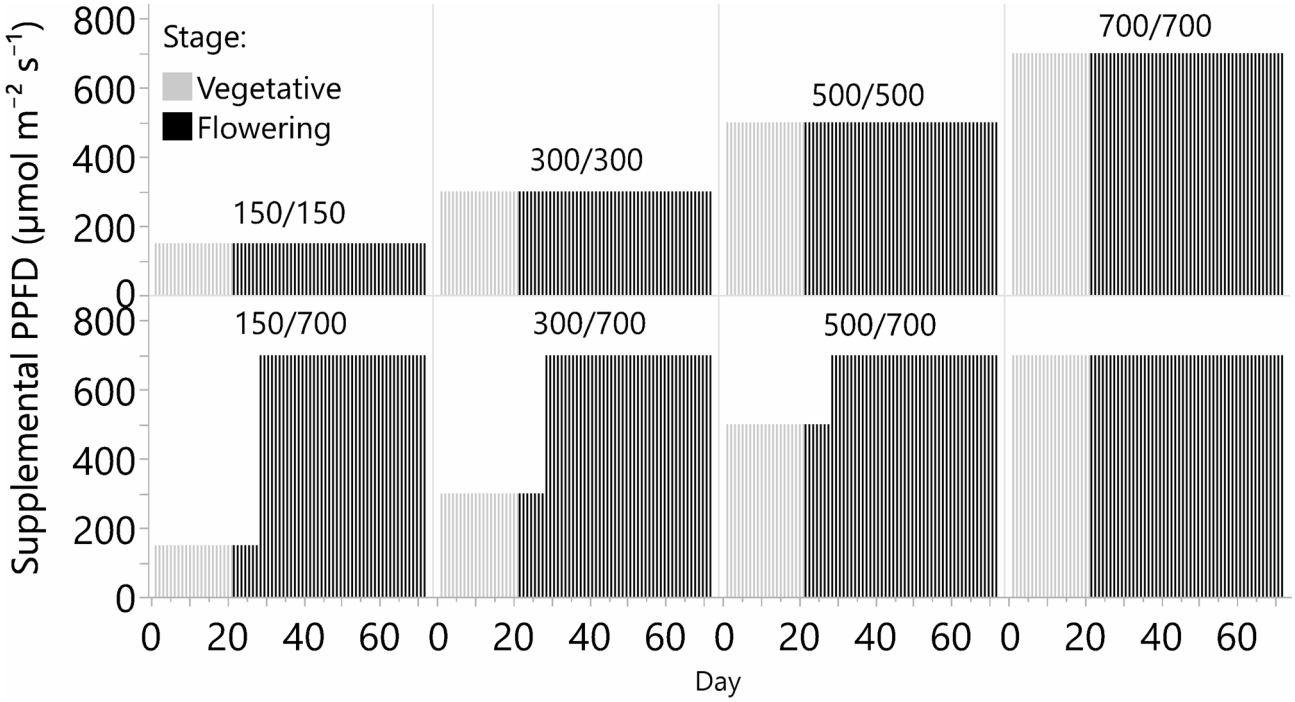


Fig. 7. Seven supplemental LED treatments. Top treatments: Static PPFD levels of 150, 300, 500, or 700 µmol m^{-2} s^{-1} for 72 days. Bottom treatments: Dynamic PPFD levels of 150, 300, or 500 µmol m^{-2} s^{-1} for 28 days, followed by 700 µmol m^{-2} s^{-1} for the final 44 days. Vegetative stage: photoperiod 18 h. Flowering stage: photoperiod 12 h.

barrier between plants; however, because two treatments were grown in each 700-PPFD plot, those plots were considered made of two subplots (Table 1).

Environmental conditions, irrigation, nutrition, and growing media

Temperature, relative humidity (RH), carbon dioxide, and solution and plant nutrition data are available in Supplementary Material Tables S1 and S2. Plants were fertirrigated automatically in each plot when substrate moisture levels were at 80% of container capacity (0.345 m^3 m^{-3}). The commercial substrate mix was a peat-based substrate (Pro-Line C/P, Jolly Gardener) evaluated by Noah et al.[79]. For details on sensors and control capabilities, refer to[7].

Water use and WUE

Evapotranspiration (ET), or water usage (WU) in this study, was assessed by the cumulative differences from day 0 to 73 in the load-cell measurements 40 min after an irrigation event and 5 min before the next event. Each irrigation lasted 25 min. ET was calculated from two load cells per plot or one load cell per subplot, with one plant per load cell. The flower production water use efficiency, simplified as flower WUE, was assessed as the ratio between the flower dry mass at 10.2% moisture and the liters of evapotranspiration in the 73-day greenhouse period. Likewise, the Leaf WUE was evaluated as the ratio between photosynthesis and transpiration, A/E.

Flower measurements and partitioning

Flowers were manually trimmed, and their fresh mass was quantified from each main stem node. Flowers were dried at ambient temperature (24.3 ± 0.2 °C) and low relative humidity (26.3 ± 0.2%) to determine commercial dry mass, resulting in this study a 10.2 ± 0.5% moisture level across treatments. Representative samples continued to dry at 70 °C to determine dry mass (0% moisture levels). All the plant inflorescences were used to calculate the total flower yields at 0 and 10.2 ± 0.5% moisture levels. In addition, commercial flower dry mass was evaluated based on the main stem's top, middle, and bottom nodes; each section accounted for one-third of the nodes.

Cannabinoids and terpenes

All flowers below the third node on the main stem were used for terpene and cannabinoid analyses. An independent laboratory (Delta 9 Analytical, Raleigh, NC) analyzed the inflorescences at 10.2% moisture. Results are based on the lab procedures and equipment available in Supplementary Material. Terpenes refer only to monoterpenes and sesquiterpenes and include terpenoids. Terpenes %, cannabinoids %, total CBD (cannabidiol) %, and total THC (delta-9-tetrahydrocannabinol) % were expressed per mass of flower (% = g/g of flowers*100). Moreover, total THC and CBD % were calculated as total THC % = THC % + 0.877*THCA (delta-9-tetrahydrocannabinolic acid) % and total CBD % = CBD % + 0.877*CBDA (cannabidiolic acid) %. In

addition, the relative percentual contribution of a single terpene or cannabinoid to the total was calculated as % = g/g of all terpenes or cannabinoids * 100.

Leaf gas exchange

Net photosynthesis (*A*), stomatal conductance (g_{sw}), and transpiration (*E*) were measured following the same protocol and conditions as on day 19[7]. The maximum leaf WUE rate was measured and estimated using fitting equations. On the other hand, the maximum photosynthetic rate (*Amax*) was estimated from the light response curve fitting equations[31,80].

Statistics

Following up the vegetative growth period in a randomized design[7], the second part of the experiment was rearranged using an incomplete block design. All plants and plot light levels were rearranged to accommodate the new lighting conditions on day 29 (Table 1). Light levels of six random plots were set at 700 µmol m^{-2} s^{-1}, while each plot was statistically considered to be made of two subplots. One subplot per plot accommodated one of three replications of 150/700, 300/700, 500/700, or 700/700 (n = 3). In contrast, the six remaining plots accommodated the static 150/150, 300/300, and 500/500 treatments with two plots per treatment (n = 2). Contrary to the number of replications, the number of observational units (OU)[81] was one with n = 3 and two with n = 2. Mixed models accounted for random effects, such as plots and subplots. In addition, one-sided t-tests were used based on our hypotheses to test the impact of increasing light on the increase of yield, cannabinoids, terpenes, and efficiency parameters at the crop and leaf levels; otherwise, a two-sided t-test was computed. For leaf *E* and WUE, a split-plot design was used to analyze the effects of ∑PPFD at the whole plot level (greenhouse plots) and short-term PPFD at the split-plot level (repeated measurements on the same leaf or plot). For *E* and Leaf WUE, the time of the day was included in the models ($p \leq 0.07$). An interaction factor significance threshold of $p > 0.20$ was used, assuming equal treatment slopes and for a generalizable interpretation of the main factor significance[82]. Data transformation was used when needed to fulfill statistical assumptions. Moreover, extreme data points were not considered for analysis, such as a single value that was notably different from other data points and caused dramatic changes in statistical significance (e.g., Fig. 1). All the analyses and graphs were done using JMP Pro 17.0 and 17.2 from SAS (Cary, NC, USA).

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

CC: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing. RH: Conceptualization, Data curation, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing.

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Competing interests

The authors declare no competing interests.

Ethics statement

All experiments involving plant material were performed in accordance with relevant national and international guidelines. Initial plant material was donated by a commercial nursery located in Broadway, North Carolina.

Additional information

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Correspondence and requests for materials should be addressed to C.E.C. or R.H.

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